

Dynamic Guideline: WHO-Based Trypanosomiasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Disease scope

“Trypanosomiasis” includes two major human diseases:

1. Human African trypanosomiasis (HAT; sleeping sickness)

- Caused by *Trypanosoma brucei gambiense* or *T. b. rhodesiense*.
- Transmitted by infected tsetse flies.
- WHO states that HAT is “almost invariably fatal unless treated” [WHO HAT treatment guideline, 2024](#).
- *T. b. gambiense* accounted for about **92% of reported HAT cases in 2018–2022** [WHO HAT treatment guideline, 2024](#).

2. American trypanosomiasis / Chagas disease

- Caused by *Trypanosoma cruzi*.
- Transmitted mainly by triatomine vectors, but also congenitally, by transfusion, transplantation, oral transmission, and accidental contamination [Towards elimination of Chagas disease, 2026](#).
- Untreated infection is lifelong, and an estimated **30–40%** of untreated people develop severe cardiac, digestive, neurological, or associated complications [Towards elimination of Chagas disease, 2026](#).

1.2 Exactly 5 WHO / PAHO Normative Guidelines or Technical Documents Identified

Important clarification: The most recent WHO item with the title pattern “**Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline**” is not related to trypanosomiasis. WHO identifies it as “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**”, launched through a WHO webinar scheduled for **11 December 2025** [WHO youth engagement event page](#). It is therefore acknowledged here for search-scope completeness but is **not counted** as a trypanosomiasis guideline.

No.	WHO / PAHO document	Publication / update date	Disease focus	Clinical relevance
1	Guidelines for the treatment of human African trypanosomiasis	28 June 2024	HAT: <i>T. b. gambiense</i> and <i>T. b. rhodesiense</i>	Latest WHO clinical treatment guideline; incorporates fexinidazole for rhodesiense HAT WHO , WHO IRIS PDF
2	WHO interim guidelines for the treatment of gambiense human African trypanosomiasis	2019	Gambiense HAT	Introduced fexinidazole into WHO HAT guidance and updated earlier therapeutic guidance WHO interim gambiense HAT guideline
3	Guidelines for the diagnosis and treatment of Chagas disease	21 November 2018	Chagas disease / <i>T. cruzi</i>	PAHO/WHO GRADE-based diagnosis and treatment guideline for adults and children WHO/PAHO
4	Research priorities for Chagas disease, human African trypanosomiasis and leishmaniasis: WHO	21 September 2012	HAT, Chagas disease, leishmaniasis	Research-priority technical report covering diagnostics, drugs, vector control, surveillance, asymptomatic infection,

No.	WHO / PAHO document	Publication / update date	Disease focus	Clinical relevance
	Technical Report Series No. 975			HIV co-infection, vaccines, and health-care interventions WHO TRS 975
5	WHO neglected tropical disease road map / criteria and procedures for elimination verification for gambiense HAT	2024–2026 reporting cycle	HAT elimination and surveillance	Normative elimination framework; WHO documents report 2024 HAT publications and verification criteria for elimination of transmission of gambiense HAT WHO NTD road map document

1.3 Latest WHO Guideline: “Guidelines for the treatment of human African trypanosomiasis”

- **Guideline title:** *Guidelines for the treatment of human African trypanosomiasis*
- **Publication date:** 28 June 2024
- **Target population:** People with confirmed or suspected HAT due to *T. b. gambiense* or *T. b. rhodesiense*, including adults, children, and pregnant women in specified clinical circumstances.
- **Scope and objectives:**
 - Provide updated treatment recommendations for both **gambiense HAT** and **rhodesiense HAT**.
 - Incorporate expanded use of **fexinidazole**, especially for **rhodesiense HAT**, substantially changing prior therapeutic pathways [WHO](#).
 - Reduce reliance on more toxic or logistically difficult regimens such as melarsoprol, suramin, pentamidine injections, and nifurtimox–eflornithine combination therapy (NECT) where appropriate [WHO IRIS PDF](#).

1.4 Guideline Development Methodology

The 2024 WHO HAT guideline:

- Used the **GRADE approach** to formulate recommendations [WHO IRIS PDF](#).
- Considered:
 - Balance of desirable and undesirable effects.
 - Certainty of evidence.
 - Resource implications.
 - Equity.
 - Acceptability.
 - Feasibility [NCBI Bookshelf executive summary](#).
- Was reviewed by the WHO guideline development group, WHO steering committee, methodologist, peer reviewers, and WHO Guidelines Review Committee [NCBI Bookshelf executive summary](#).
- Used methodology consistent with the 2019 WHO interim gambiense HAT guideline [WHO IRIS PDF](#).

GRADE framework used in this report

GRADE category	Interpretation
High	Further research is very unlikely to change confidence in the estimate of effect.
Moderate	Further research may affect confidence and may change the estimate.
Low	Further research is likely to affect confidence and may change the estimate.
Very low	Any estimate of effect is very uncertain.
Strong recommendation	Most informed patients should receive the intervention; clinicians can structure practice accordingly.
Conditional recommendation	Shared decision-making and context-specific implementation are important.

1.5 Key WHO-Based Recommendations for Human African Trypanosomiasis

A. Rhodesiense HAT

Clinical scenario	WHO-based recommendation	GRADE evidence quality	Strength
First-stage rhodesiense HAT , age ≥6 years and weight ≥20 kg	Use oral fexinidazole rather than suramin	Very low	Conditional
Second-stage rhodesiense HAT , age ≥6 years and weight ≥20 kg	Use oral fexinidazole rather than melarsoprol	Very low	Conditional
Second-stage rhodesiense HAT where oral therapy is not reliable	Prefer melarsoprol if the patient cannot swallow, has persistent vomiting, has contraindication to fexinidazole, or oral absorption is questionable	Very low	Conditional
Rhodesiense HAT, age <6 years or weight <20 kg, first-stage disease	Use suramin	Very low	Conditional
Rhodesiense HAT, age <6 years or weight <20 kg, second-stage disease	Use melarsoprol	Very low	Conditional
Suspected rhodesiense HAT when first-choice drugs are not immediately available	Consider pentamidine as immediate interim therapy while awaiting recommended drugs	Very low	Conditional

Sources: [WHO](#), [NCBI Bookshelf executive summary](#), [CDC clinical care for HAT](#), [InfoNTD summary](#).

B. Gambiense HAT

Clinical scenario	WHO-based recommendation	GRADE evidence quality	Strength
First-stage gambiense HAT , age ≥ 6 years and weight ≥ 20 kg	Use fexinidazole rather than pentamidine	Very low	Conditional
Non-severe second-stage gambiense HAT , age ≥ 6 years and weight ≥ 20 kg, CSF WBC $< 100/\mu\text{L}$	Use fexinidazole rather than NECT	Low	Conditional
Severe second-stage gambiense HAT , CSF WBC $\geq 100/\mu\text{L}$	Use NECT rather than fexinidazole	Low	Conditional
Gambiense HAT, age < 6 years or weight < 20 kg, first-stage disease	Use pentamidine	Very low	Conditional
Gambiense HAT, age < 6 years or weight < 20 kg, second-stage disease	Use NECT	Very low	Conditional
Pregnant women with gambiense HAT	Follow former WHO recommendations; fexinidazole may be given after the first trimester when appropriate	Very low	Conditional

Sources: [WHO IRIS PDF](#), [NCBI Bookshelf executive summary](#), [CDC clinical care for HAT](#).

1.6 Practical Administration and Safety Principles

Area	Recommendation / implementation point	GRADE evidence quality	Strength
Fexinidazole adherence	Ensure the full 10-day course is completed.	Very low	Conditional
Food requirement	Fexinidazole should be swallowed after appropriate food intake; if this cannot be ensured, use an alternative regimen.	Very low	Conditional
Rhodesiense HAT dosing, age ≥6 years and weight ≥20 kg	CDC summarizes WHO-based dosing as: weight >20 to <35 kg: 1,200 mg daily days 1–4, then 600 mg daily days 5–10; weight >35 kg: 1,800 mg daily days 1–4, then 1,200 mg daily days 5–10.	Very low	Conditional
Severe disease or absorption concerns	Use injectable alternatives when vomiting, swallowing difficulty, or absorption concerns make oral fexinidazole unreliable.	Very low	Conditional

Sources: [WHO](#), [CDC clinical care for HAT](#).

1.7 Diagnosis and Staging: HAT

Diagnostic action	Clinical use	GRADE evidence quality	Strength
Consider HAT in compatible illness plus exposure to endemic tsetse regions	Applies to residents, migrants, expatriates, and travelers	Low	Strong
Confirm infection by finding trypanosomes in blood, lymph-node aspirate, chancre fluid, or CSF when possible	Core diagnostic standard	Low	Strong

Diagnostic action	Clinical use	GRADE evidence quality	Strength
Use buffy coat, microhematocrit centrifugation, mini-anion exchange centrifugation, or serial microscopy to improve sensitivity	Especially useful when parasitemia is low	Low	Conditional
Use CATT for gambiense HAT population screening in endemic areas, but not as confirmatory diagnosis	CATT is insufficiently specific for confirmation and is not available in the United States	Low	Conditional
Do not use serology for rhodesiense HAT diagnosis	CDC states serologic testing is not used for <i>T. b. rhodesiense</i> diagnosis	Low	Strong
Perform lumbar puncture selectively when treatment choice depends on stage	Less necessary in eligible patients treated with fexinidazole for both stages; still relevant in children below fexinidazole thresholds and severe gambiense HAT	Very low	Conditional

Sources: [CDC clinical guidance for HAT](#), [MSD Manual](#), [Frontiers review](#).

1.8 WHO / PAHO-Based Recommendations for Chagas Disease

The most recent WHO/PAHO Chagas diagnosis and treatment guideline identified was published on **21 November 2018** [WHO/PAHO](#). It was developed according to the WHO handbook for guideline development, used searches in PubMed, EMBASE, Cochrane, and manual sources up to **August 2017**, and used GRADE evidence profiles [WHO/PAHO](#).

Clinical scenario	WHO / PAHO / CDC-aligned recommendation	GRADE evidence quality	Strength
Acute Chagas disease	Treat with trypanocidal therapy; benznidazole or nifurtimox are principal options	Low	Strong
Congenital Chagas disease	Treat as acute disease	Low	Strong
Chronic <i>T. cruzi</i> infection in children	Treat chronic infection in children and adolescents; CDC indicates treatment up to age 18	Low	Strong
Adults ≤50 years with chronic infection and no advanced cardiomyopathy	Offer / strongly recommend antiparasitic treatment, weighing clinical status	Low	Strong
Adults >50 years with chronic infection	Individualize treatment after weighing risks, benefits, comorbidities, and patient preference	Low	Conditional
Acute or chronic Chagas disease when trypanocidal therapy is prescribed	Use either benznidazole or nifurtimox	Low	Conditional
Pregnancy	Avoid benznidazole and nifurtimox during pregnancy; consider treatment after pregnancy and breastfeeding when appropriate	Low	Strong
Severe hepatic or renal disease	Avoid treatment or seek expert consultation due to contraindications	Low	Strong
Advanced Chagas cardiomyopathy	Antiparasitic treatment is generally not recommended in advanced cardiomyopathy	Low	Conditional

Sources: [WHO/PAHO Chagas guideline](#), [PAHO guideline PDF](#), [WHO Chagas fact sheet](#), [CDC Chagas clinical care](#).

1.9 Current WHO Epidemiological and Elimination Context

Indicator	WHO-reported status	Source
Total HAT cases in 2024	583	WHO World NTD Day 2026
Gambiense HAT cases in 2024	546 cases in 10 countries	WHO NTD road map document
Rhodesiense HAT cases in 2024	37 cases in 4 countries, a 94% reduction from 2000	WHO NTD road map document
HAT elimination target	Gambiense HAT: interruption of transmission; rhodesiense HAT: elimination as a public health problem	WHO World NTD Day 2026
Countries validated as having eliminated HAT as a public health problem	WHO reported nine countries by World NTD Day 2026	WHO World NTD Day 2026
Countries with interruption of HAT transmission	None yet reported by WHO	WHO World NTD Day 2026
Chagas progress	By December 2025, eight South American countries had interrupted domiciliary vector transmission nationally or in defined areas	WHO NTD road map document

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Time window

- Latest WHO HAT guideline release: 28 June 2024.
- Evidence reviewed after release through: 29 April 2026.

Search strategy

Evidence sources prioritized:

- WHO official publications, WHO IRIS, WHO NTD pages, WHO Executive Board documents.
- CDC clinical guidance.
- PubMed / PMC-indexed studies.
- Lancet-family clinical trial publications.
- Clinical trial registry identifiers where available.
- Regulatory updates from EMA / CHMP and major development partners.

Inclusion criteria

Included evidence if it addressed:

- HAT diagnosis, staging, treatment, drug implementation, or elimination.
- Chagas disease diagnosis, treatment, elimination, or WHO/PAHO implementation.
- New evidence or regulatory developments after **28 June 2024**, or pre-existing pivotal trials newly relevant due to 2026 regulatory decisions.

Exclusion criteria

Excluded or downgraded evidence if:

- It was not human trypanosomiasis-related.
- It lacked extractable clinical findings.
- It was a narrative or policy statement without primary data.
- It was not applicable to diagnosis, treatment, management, or elimination.

2.2 Key New Studies and Evidence After 28 June 2024

Study / evidence item	Design	Population / intervention	Key findings	GRADE quality
Matovu et al., <i>Lancet Global Health</i> 2025: fexinidazole for rhodesiense HAT	Prospective, open-label, single-arm phase 2–3, non-randomized study; NCT03974178	Patients ≥6 years and ≥20 kg with parasitologically confirmed rhodesiense HAT; 10-day oral fexinidazole	Authors concluded fexinidazole was safe, easy to use, and a better-accepted alternative to melarsoprol or suramin PubMed	Low
2025 PubMed-indexed HAT review	Narrative clinical review	HAT diagnosis, treatment, and control	Stated HAT control relies on case detection, treatment, and vector control; rapid diagnostic tests improved gambiense screening; fexinidazole is first-line for gambiense HAT without severe meningoencephalitic disease and for rhodesiense HAT PubMed	Very low
Acoziborole phase 2/3 trial, <i>Lancet Infectious Diseases</i> / PMC	Multicentre, open-label, single-arm phase 2/3 study	Gambiense HAT; single-dose oral acoziborole	Treatment success at 18 months was 95.2% : 159/167 patients; 95% CI 91.2–97.7 PMC	Low
Acoziborole safety data from trial report	Single-arm phase 2/3 safety dataset	208 participants	169/208 had ≥1 adverse event; 155/208 had ≥1 treatment-emergent	Low

Study / evidence item	Design	Population / intervention	Key findings	GRADE quality
			adverse event; 28/208 had ≥1 severe treatment-emergent adverse event; no severe or serious drug-related treatment-emergent adverse events were reported in the summarized table PMC	
EMA / CHMP positive opinion for Acoziborole Winthrop, 2026	Regulatory review	Adults and adolescents with gambiense HAT meeting specified criteria	Acoziborole described as a one- day / three-tablet single-dose oral treatment; CHMP positive opinion reported in 2026 Sanofi , Nature	Low
EDCTP / STROGHAT implementation update, 2026	Implementation research / programmatic screening	At-risk populations in gambiense HAT areas	454,871 screening tests over two years; 95% at-risk population coverage; 0.43% seropositivity; 803 eligible individuals treated with single-dose acoziborole by end- 2025 Global Health EDCTP3	Very low
WHO World NTD Day 2026 HAT update	WHO surveillance / programme report	Global HAT programme	583 total HAT cases in 2024; nine countries validated as having eliminated	High for surveillance; not

Study / evidence item	Design	Population / intervention	Key findings	GRADE quality
			HAT as a public health problem; no country had interrupted transmission WHO World NTD Day 2026	intervention GRADE
2026 Bulletin perspective on Chagas elimination	WHO Bulletin perspective	Chagas disease elimination policy	Emphasized political commitment, simplified tools, equitable treatment access, primary-care integration, community engagement, and housing improvements Towards elimination of Chagas disease, 2026	Very low

2.3 Interpretation of New Evidence

A. Fexinidazole for rhodesiense HAT

The 2025 *Lancet Global Health* rhodesiense HAT study provides post-guideline clinical support for the WHO 2024 decision to recommend fexinidazole as first-line therapy in eligible patients. However:

- It was **open-label** and **single-arm**.
- It used historical expectations rather than a concurrent randomized comparator.
- Rhodesiense HAT is rare, making large randomized trials difficult.

Guideline impact: Supports the 2024 WHO recommendation but does **not** justify changing conditional recommendations to strong recommendations.

B. Fexinidazole for gambiense HAT

Evidence remains consistent with WHO's position:

- Fexinidazole is appropriate for eligible first-stage and non-severe second-stage gambiense HAT.
- **NECT remains preferred** for severe second-stage gambiense HAT with CSF WBC $\geq 100/\mu\text{L}$.

A Cochrane-style review summarized in PMC found that, in second-stage gambiense HAT, mortality at 24 months **may be higher** with fexinidazole, relapse probably increases, and the effect on serious adverse events is uncertain compared with NECT [PMC](#). This supports WHO's caution in severe second-stage disease.

C. Acoziborole for gambiense HAT

Acoziborole is the most important emerging therapy after the 2024 WHO guideline:

- Single-dose oral treatment.
- 18-month treatment success of **95.2%** in a phase 2/3 trial [PMC](#).
- Positive CHMP opinion reported in 2026 [Sanofi](#).
- Large-scale implementation research underway, with 803 eligible individuals treated by end-2025 [Global Health EDCTP3](#).

Guideline impact: Promising but should not replace WHO recommendations until WHO issues updated guidance or countries adopt it through regulatory and programme channels.

D. Chagas disease

No new WHO/PAHO clinical treatment guideline replacing the 2018 Chagas guideline was identified. The 2026 WHO Bulletin perspective emphasizes elimination feasibility through:

- Primary-care integration.
- Community engagement.
- Equitable access to treatment.
- Simplified tools.
- Housing and structural interventions [Towards elimination of Chagas disease, 2026](#).

Guideline impact: No change to core benznidazole / nifurtimox recommendations.

2.4 Updated Recommendations Based on New Evidence

Original WHO recommendation	New evidence impact through 29 April 2026	Updated recommendation	GRADE quality	Strength
Use fexinidazole for eligible first-stage rhodesiense HAT patients ≥ 6 years and ≥ 20 kg	2025 phase 2–3 evidence supports safety and feasibility	Maintain recommendation: fexinidazole remains first-line in eligible patients	Low	Conditional
Use fexinidazole for eligible second-stage rhodesiense HAT patients ≥ 6 years and ≥ 20 kg	New study supports avoiding melarsoprol in many patients, but evidence remains non-randomized	Maintain recommendation; use melarsoprol if oral therapy is not reliable	Low	Conditional
Use fexinidazole for first-stage and non-severe second-stage gambiense HAT in eligible patients	No post-guideline evidence overturns this; existing comparative concerns remain for severe disease	Maintain recommendation for eligible non-severe disease	Low	Conditional
Use NECT for severe second-stage gambiense HAT	Comparative evidence supports caution with fexinidazole in severe second-stage disease	Maintain recommendation: NECT preferred when CSF WBC $\geq 100/\mu\text{L}$	Low	Conditional
Use pentamidine / NECT in gambiense HAT children < 6 years or < 20 kg	No new evidence supports change	Maintain recommendation	Very low	Conditional

Original WHO recommendation	New evidence impact through 29 April 2026	Updated recommendation	GRADE quality	Strength
Use suramin / melarsoprol in rhodesiense HAT children <6 years or <20 kg	No new evidence supports replacing these regimens in this group	Maintain recommendation	Very low	Conditional
No WHO recommendation yet for acoziborole in routine HAT treatment	Acoziborole has strong programmatic promise and 2026 CHMP positive opinion	Do not replace WHO regimens yet; consider only where approved and implemented under national/WHO-aligned protocols	Low	Conditional
Treat Chagas disease with benznidazole or nifurtimox when trypanocidal therapy is indicated	No new WHO/PAHO clinical replacement guideline found	Maintain 2018 PAHO/WHO recommendation	Low	Conditional

2.5 Management Algorithm: Practical Clinical Approach

A. Suspected HAT

1. Assess exposure

- Residence or travel in tsetse-endemic sub-Saharan Africa.
- Compatible fever, lymphadenopathy, chancre, sleep disturbance, neurological symptoms.

2. Identify likely subspecies

- West / Central Africa: usually *T. b. gambiense*.
- East / Southern Africa: usually *T. b. rhodesiense*.

3. Confirm parasitologically when possible

- Blood microscopy, buffy coat, chancre fluid, lymph-node aspirate, or CSF microscopy [CDC clinical guidance for HAT](#).

4. Stage disease when treatment choice requires it

- Lumbar puncture is less necessary in eligible rhodesiense HAT patients treated with fexinidazole but remains important in specific pediatric and severe gambiense HAT scenarios.

5. **Treat urgently**

- HAT is generally fatal without treatment [WHO HAT treatment guideline, 2024](#).

B. Suspected Chagas Disease

1. **Assess risk**

- Birth, residence, maternal exposure, transfusion, transplant, or vector exposure in endemic areas.
- Consider congenital infection in infants of infected mothers.

2. **Confirm diagnosis**

- Acute infection: direct parasite detection may be possible.
- Chronic infection: serologic testing using appropriate assays; follow local or national algorithms.

3. **Treat priority groups**

- Acute, congenital, and reactivated infection.
- Children and adolescents.
- Adults ≤ 50 years without advanced cardiomyopathy.
- Individualized decisions for adults > 50 years.

4. **Avoid treatment or obtain specialist input**

- Pregnancy.
- Breastfeeding.
- Severe renal or hepatic disease.
- Advanced Chagas cardiomyopathy.

Sources: [WHO/PAHO Chagas guideline](#), [WHO Chagas fact sheet](#), [CDC Chagas clinical care](#).

2.6 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority
Randomized comparative evidence for rhodesiense HAT treatment	Current fexinidazole evidence is encouraging but limited by rarity and single-arm designs	High
Optimal treatment for severe second-stage gambiense HAT	NECT remains preferred, but simpler regimens are needed	High

Evidence gap	Why it matters	Priority
Role of acoziborole in WHO treatment algorithms	Single-dose therapy could transform HAT elimination strategies	High
Pediatric acoziborole data	Current regulatory discussions emphasize adults/adolescents meeting age and weight criteria	High
Use of oral regimens in pregnancy	Evidence remains very limited	High
Simplified diagnostics for low-prevalence HAT settings	Elimination requires sensitive surveillance and confirmation tools	High
Chagas disease treatment in older adults and advanced cardiomyopathy	Benefit-risk balance remains uncertain	High
Chagas congenital screening and treatment implementation	Early treatment is highly effective, but access and diagnosis gaps remain	High
Integration of vector control, housing, and primary care for Chagas disease	WHO emphasizes structural determinants and primary-care integration	Moderate
Drug resistance monitoring for fexinidazole, nifurtimox, and related nitroimidazoles	Cross-resistance mechanisms are biologically plausible and require surveillance	Moderate

Sources: [WHO TRS 975](#), [Frontiers review](#), [Towards elimination of Chagas disease, 2026](#).

2.7 Final Dynamic Guideline Position as of 29 April 2026

1. **For HAT, the 2024 WHO treatment guideline remains the controlling WHO clinical guideline.**
2. **Fexinidazole is central to current HAT treatment**, especially:
 - Eligible rhodesiense HAT patients ≥ 6 years and ≥ 20 kg.
 - Eligible gambiense HAT patients with first-stage or non-severe second-stage disease.
3. **NECT remains preferred for severe second-stage gambiense HAT.**
4. **Melarsoprol and suramin remain necessary**, particularly in younger or lower-weight rhodesiense HAT patients and when oral fexinidazole cannot be used.
5. **Acoziborole is the major emerging therapy** for gambiense HAT, with 2026 CHMP positive opinion and promising phase 2/3 results, but it should not replace WHO regimens until

incorporated into WHO or national guidelines.

6. **For Chagas disease, the 2018 PAHO/WHO guideline remains the latest identified WHO/PAHO clinical guideline.**
7. **Benznidazole and nifurtimox remain the core antiparasitic treatments** for Chagas disease, with strongest priority for acute, congenital, reactivated, pediatric, and selected adult chronic infection.
8. **Elimination efforts must combine diagnosis, treatment, surveillance, vector control, primary-care integration, and equitable access**, consistent with WHO NTD strategy and 2026 WHO elimination perspectives.